

Gravitational-Wave Scattering off a Thin-Shell Vacuum Bubble

Part I: Israel Junction Conditions and the Background Dynamics

Goal. A thin spherical shell (domain wall) separates flat interior ($r < R$) from Schwarzschild exterior ($r > R$, mass M). We want the reflection and transmission coefficients for a gravitational wave incident from infinity, scattering off the shell, with the interior solution regular at $r = 0$ (constructed previously) and the exterior carrying both an incoming and an outgoing mode. This requires (i) the linearized Israel junction conditions relating the jump in the perturbed extrinsic curvature to the perturbed surface stress-energy of the shell, and (ii) a background (unperturbed) configuration to linearize around.

Status of this document. Part I below is complete and self-contained: the general thin-shell formalism, the vacuum-bubble (domain-wall) stress tensor, and — critically — the background dynamics of a spherical domain wall, which turns out *not* to admit a static configuration. This changes the shape of the scattering problem and is why Part II (the actual linearized matching and the R, T formulas) is deferred to a follow-up once we agree on how to handle this. See the discussion in Sec. 5 and the question at the end.

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1 General thin-shell (Gauss–Codazzi–Israel) formalism

We follow Ipser & Sikivie [1], who in turn follow Israel [2]. Let Σ be a timelike hypersurface with unit *outward* normal ξ^α ($\xi_\alpha \xi^\alpha = +1$), induced metric $h_{\alpha\beta} = g_{\alpha\beta} - \xi_\alpha \xi_\beta$, and extrinsic curvature

$$\pi_{ab} := D_a \xi_b = \pi_{ba}, \quad D_a := h_a^b \nabla_b \quad (1)$$

(lower-case Latin indices a, b run over the three coordinates intrinsic to Σ). If the bulk stress-energy $T_{\alpha\beta}$ has a δ -function singularity on Σ , one defines the jump and average

$$\gamma_{ab} := \pi_{ab}|_+ - \pi_{ab}|_-, \quad \tilde{\pi}_{ab} := \frac{1}{2}(\pi_{ab}|_+ + \pi_{ab}|_-), \quad (2)$$

($\pm$ = just outside/inside Σ along ξ^α), and the surface stress-energy $S_{ab} := \int T_{ab} d\ell$ (ℓ = proper distance across Σ). The junction condition (Ipser & Sikivie’s Eq. (2.6), citing Israel [2]) is

$$S_{ab} = -\frac{1}{8\pi}(\gamma_{ab} - h_{ab}\gamma^c_c), \quad \text{i.e.} \quad [\pi_{ab}] - h_{ab}[\pi] = -8\pi S_{ab}, \quad (3)$$

together with continuity of the induced metric h_{ab} across Σ (the “first” junction condition). This — plus the vacuum field equations off Σ — is the complete set of matching conditions.

2 The vacuum-bubble (domain-wall) stress tensor

Following Zel’dovich, Kobzarev & Okun (as quoted in [1]) and Ipser & Sikivie’s Eq. (2.9), the most general isotropic surface stress tensor is

$$S^{ab} = \sigma u^a u^b - \tau(h^{ab} + u^a u^b), \quad (4)$$

with u^a the four-velocity of an observer comoving with Σ , σ the surface energy density, and τ the surface tension. A *domain wall* (a bubble wall separating two vacua) has $\tau = \sigma$ [1], giving the remarkably simple

$$\boxed{S^{ab} = -\sigma h^{ab}} \quad (\text{Ipser \& Sikivie, Eq. (2.10)}), \quad (5)$$

i.e. the surface stress tensor is exactly proportional to the induced metric — a Nambu-Goto-type membrane of fixed tension σ , with *no internal degrees of freedom beyond its own position*. Ipser & Sikivie further show, from the conservation identity $D_b S^{ab} = 0$, that for $\tau = \sigma$, “ σ is a constant, i.e., σ has the same value at all events on the three-dimensional timelike surface S ” [1]. This is the key structural fact we will use later: **σ is not an independent dynamical field that can be perturbed** — the wall’s only perturbation is a perturbation of its own embedding/induced metric, which already appears on both sides of Eq. (5).

3 Spherical domain wall: background configuration

Interior flat, exterior Schwarzschild (Birkhoff, as in our earlier interior-wave note), shell at $r = R(t)$:

$$(ds^2)_+ = -f_+ dt^2 + f_+^{-1} dr^2 + r^2 d\Omega^2, \quad f_+ = 1 - \frac{2M}{r} \quad (r > R), \quad (ds^2)_- = -dT^2 + dr^2 + r^2 d\Omega^2 \quad (r < R), \quad (6)$$

[1], Eqs. (3.1). With $\alpha := \dot{T} = (1 + \dot{R}^2)^{1/2}$, $\beta := (f_+ + \dot{R}^2)^{1/2}$ (dot = $d/d\tau$, proper time along the wall), the angular part of the extrinsic curvature is

$$K^\theta_\theta|_+ = \frac{\beta}{R}, \quad K^\theta_\theta|_- = \frac{\alpha}{R} \quad (\text{exterior, interior, respectively}), \quad (7)$$

matching Ipser & Sikivie’s Eq. (3.5)-(3.6) exactly, and substitution of Eq. (5) into the junction condition (3) yields their equations of motion, Eq. (3.7):

$$(\alpha + \beta)\ddot{R} = -\frac{\alpha M}{R^2} - \frac{2\tau}{\sigma} \frac{\alpha\beta(\alpha + \beta)}{R}, \quad (\alpha - \beta)\ddot{R} = -\frac{\alpha M}{R^2} + 4\pi(\sigma - 2\tau)\alpha\beta. \quad (8)$$

3.1 Key result: a pure-tension bubble never sits still

Setting $\tau = \sigma$ (domain wall) in the first of Eq. (8): since $\alpha, \beta, M/R^2 > 0$ and $\tau/\sigma = 1$, the right-hand side is manifestly negative, so $\ddot{R} < 0$ always. Quoting Ipser & Sikivie directly: “*Equation (3.7a) tells us that $\ddot{R}$ is always negative if $\tau \geq 0$. Hence, a spherical wall with $\tau \geq 0$ always collapses to a black hole,*

regardless of its size” [1]. **There is no static equilibrium radius for a pure-tension vacuum bubble.** (One can check this directly too: setting $\dot{R} = \ddot{R} = 0$ in Eq. (8) with $\tau = \sigma$ forces $M/R^2 = 0$, i.e. $M = 0$ — the trivial, gravity-free case.)

Eliminating $\ddot{R}$ from Eq. (8) gives the mass formula, Eq. (3.8):

$$M = \frac{1}{2} \left[(1 + \dot{R}^2)^{1/2} + (1 - 2M/R + \dot{R}^2)^{1/2} \right] 4\pi\sigma R^2, \quad (9)$$

an implicit first integral of the motion (using $\dot{M} = 0$, Birkhoff, and $\dot{\sigma} = 0$). Setting $\dot{R} = 0$ (the turning point of the collapse/bounce trajectory — the bubble’s radius of maximum expansion, R_m) gives Ipser & Sikivie’s Eq. (3.9),

$$M = 4\pi\sigma R_m^2 (1 - 2\pi\sigma R_m), \quad R_m < \frac{1}{4\pi\sigma}. \quad (10)$$

This is a genuine, unambiguous, momentarily-static snapshot of the geometry: given a wall tension σ and a chosen R_m , Eq. (10) fixes M , and at that instant $\dot{R} = 0$ (though $\ddot{R} < 0$, so the bubble is about to start collapsing).

4 Consequence for the scattering problem, and a proposed way forward

A stationary (single-frequency, time-independent-background) scattering calculation of the kind we set up for the interior wave equation presupposes a *static* boundary at $r = R$. Section 3.1 shows this is impossible for an idealized pure-tension domain wall with nonzero M : the shell is intrinsically dynamical (it collapses). Three ways to proceed, in increasing order of departure from “the vacuum bubble, exactly as in Ipser & Sikivie”:

1. **Adiabatic / high-frequency approximation.** Freeze the background at the turning point $R = R_m$, $\dot{R} = 0$ [Eq. (10)], and compute $R(\omega), T(\omega)$ as if the shell were momentarily at rest there — valid when the gravitational wave’s period is much shorter than the collapse timescale, $\omega \gg |\ddot{R}/\dot{R}|$ near R_m (formally infinite right at the turning point, so more precisely $\omega \times (\text{wave-crossing time of the shell region}) \gg 1$, the usual eikonal/short-wavelength condition). This keeps $\dot{R} = 0$ but *not* $\ddot{R} = 0$, so the background K^τ_τ jump retains an $\ddot{R}$ -dependent piece that must be kept in the linearized matching (it feeds into the trace term $h_{ab}[\pi]$ of Eq. (3) at the perturbed level).
2. **Relax the pure-tension condition.** Allow $\tau \neq \sigma$ (not a strict domain wall, but a general isotropic shell) and look for a genuine static solution of Eq. (8) ($\dot{R} = \ddot{R} = 0$ for *all* time, i.e. a true equilibrium, not just a turning point). This exists for special combinations of τ/σ , M , R (Ipser & Sikivie discuss the analogous planar case in their Sec. IV B, finding static solutions require $\tau = \sigma$ or $\tau = \frac{1}{4}\sigma$ for the planar wall [1]; the spherical analogue would need to be worked out). This sacrifices the “pure vacuum bubble” character of the shell.
3. **Fully dynamical scattering.** Keep $R(t)$ evolving self-consistently (collapsing) and treat the wave scattering off a genuinely moving/accelerating boundary — a much harder, intrinsically time-dependent problem (no fixed ω), likely requiring numerical evolution or a slow-motion (multiple-scales) expansion in $\dot{R}$.

Option 1 is the natural “next step” that stays closest to the literal vacuum-bubble setup and still yields clean reflection/transmission *coefficients* (functions of ω, l) rather than a fully time-dependent scattering problem. **We recommend this and will use it unless told otherwise.**

5 Setting up the linearized matching (odd parity): what is done and what remains

This section records the technical progress made towards the linearized (perturbed) junction condition, to be completed in Part II.

5.1 Why the domain wall adds no new degrees of freedom (odd parity)

Because $S^{ab} = -\sigma h^{ab}$ with σ exactly constant (Sec. 2), the *perturbation* of the surface stress tensor is entirely determined by the perturbation of the induced metric:

$$\delta S_{ab} = -\sigma \delta h_{ab}. \quad (11)$$

There is no independent “surface fluid” variable to solve for (no surface velocity perturbation, no density perturbation) — unlike, e.g., a shell of ordinary matter with its own equation of state. This is what makes the odd-parity sector tractable as a genuinely closed linear problem.

Odd-parity perturbations, in Regge-Wheeler gauge, do not displace the shell radially (recall from our interior note: the odd-parity embedding function vanishes identically), so Σ remains exactly at $r = R$. The bulk odd-parity metric perturbation contributes $h_{tA} = h_t^{lm}(t, r) X_A^{lm}(\theta, \phi)$ and $h_{rA} = h_r^{lm}(t, r) X_A^{lm}$ (using Martel & Poisson’s [3] notation, as in our formulario), with h_{tA} (but not h_{rA} , a normal-direction component) entering the induced metric on Σ .

5.2 The perturbed extrinsic curvature (K_{tA} component)

Using the ADM decomposition with r as the foliation parameter — lapse $N = f^{-1/2}$ (unperturbed for odd parity, since g_{rr} is not perturbed), shift $N_A = h_r^{lm} X_A^{lm}$ — the odd-parity extrinsic-curvature perturbation is

$$\delta K_{tA} = \frac{1}{2N} (\partial_r h_{tA} - \partial_t N_A) = \frac{\sqrt{f}}{2} (\partial_r h_t^{lm} - \partial_t h_r^{lm}) X_A^{lm}, \quad (12)$$

(the background connection of the $r = \text{const}$ 3-slice is block-diagonal to the order needed, so no extra Christoffel corrections survive at linear order — this was checked explicitly).

5.3 What remains

The linearized tA -component of Eq. (3), using Eqs. (11) and (12), reads schematically

$$[\delta K_{tA}] - \delta h_{tA} [K]^{(0)} = 8\pi\sigma \delta h_{tA} \Big|_{r=R}, \quad (13)$$

where $[K]^{(0)} = [\pi^c_c]$ is the *background* trace jump — nonzero here (Sec. 3), which is precisely the complication introduced by the lack of a true static background: at a genuine static equilibrium this cross-term vanishes trivially, but at the turning point R_m it does not (it involves $\ddot{R} \neq 0$ through the background K^τ_τ). Completing this requires: (i) the background $[K]^{(0)}$ evaluated at R_m from Eq. (8), (ii) relating $h_t^{lm}(t, R)$ on each side to the master function Ψ_{odd} and its derivative (via Martel & Poisson’s Eq. (5.7), (5.13) [3]), and (iii) reducing the result to a clean jump/continuity statement purely in terms of $\Psi_{\text{odd}}(R^\pm)$ and $\Psi'_{\text{odd}}(R^\pm)$, which is what actually feeds the reflection/transmission linear system. This algebra is nontrivial (it must be done on the curved, $M \neq 0$ exterior side, where the $\square$ operator acting on the covector $\tilde{h}_a$ picks up genuine curvature corrections) and is deferred to Part II.

References

- [1] J. Ipser and P. Sikivie, “Gravitationally repulsive domain wall,” *Phys. Rev. D* **30**, 712 (1984). (Received 7 November 1983 — the paper is sometimes informally dated 1983 for that reason, but the publication itself is *Phys. Rev. D* **30** (1984).) Equations and quotes used here: Eq. (2.1)–(2.14) [general Gauss-Codazzi/Israel formalism and the domain-wall stress tensor], Eq. (3.1)–(3.9) and the discussion following Eq. (3.7a) [spherical domain wall dynamics and the collapse result], Sec. IV B [planar-wall static solutions, for comparison].
- [2] W. Israel, “Singular hypersurfaces and thin shells in general relativity,” *Nuovo Cimento B* **44**, 1 (1966); as cited in [1].
- [3] K. Martel and E. Poisson, “Gravitational perturbations of the Schwarzschild spacetime: A practical covariant and gauge-invariant formalism,” *Phys. Rev. D* **71**, 104003 (2005).